

## Maximising AI for driver attention and vehicle safety

Artificial Intelligence (AI) is revolutionising vehicle safety by improving driver attention and reaction to road conditions. It offers nuanced insights into driver behaviour, enhancing overall road safety.

**Understanding driver reactions.** AI decodes drivers' responses to various road situations, including emotional states and adherence to traffic rules.

**Addressing unexpected obstacles.** AI distinguishes between momentary distractions and emergencies, refining its approach to driver attention.

**Real-time collision detection.** AI predicts potential collisions by analysing data continuously, enhancing proactive safety measures.

**Key factors in risk assessment.** AI assesses variables like vehicle speed, obstacle proximity, and driver alertness for accurate risk evaluation.

**Versatile adaptation.** AI adapts across scenarios, from lane-keeping assistance to using drones for traffic condition monitoring.

The implementation of AI in vehicular systems marks a significant stride in automotive technology, drastically improving safety and driver awareness. Through real-time analysis and adaptive responses, AI is setting new benchmarks in vehicle safety, making driving a more secure and responsive experience.

is adaptable to the existing systems within an organisation, making Bobby Car a versatile tool for various business needs.

A notable application of Bobby Car is its air interface, equipped with a feature known as 'Dark Spot.' This involves a unique interaction with unit drivers, adding another layer of complexity and utility to the project. Furthermore, the geography console of Bobby Car not only handles deployment zones but also incorporates location-based services, utilising latitude and longitude for precise coordination.

## Bridging software and hardware in connected cars

Bobby Car stands out for its ability to seamlessly interact with IoT devices and hardware. This project, completed in collaboration with various organisations, including BMW and ECS, has pioneered a pattern for creating connected cars. The essence of Bobby Car is in its integration of software with the physical components of a car, embodying the intersection of digital and mechanical engineering.

The Bobby Car project is centred around three essential elements. First, it incorporates sensors that act as the project's eyes and ears, gathering environmental data. Next is the software interaction aspect, where

Bobby Car's software is specifically designed to efficiently interact with these sensors, ensuring effective data processing and utilisation. Finally, the project is grounded in its robust infrastructure, which forms the backbone by supporting and running both the software and the sensors seamlessly.

## Key technologies powering Bobby Car

Bobby Car utilises several core technologies to enhance its functionality:

**OpenShift platform.** The platform is pivotal to the project, allowing the software to efficiently scale from managing a few IoT devices or cars to potentially thousands or millions. It guarantees uniform performance across diverse deployment environments, including on-premises and cloud settings, thereby ensuring a smooth transition across these platforms. This consistency and scalability are fundamental to the project's success across different scenarios.

**Motor vehicle simulator and MQTT protocol.** These are used for real-time data transmission. The MQTT protocol, in particular, is crucial for its reliability even in unstable network conditions, making it ideal for IoT devices that require consistent connectivity.

**BAFTA messaging protocol.** This complements MQTT by providing a robust messaging infrastructure capable of handling large volumes of data from thousands of IoT devices.

**Data storage and integration.** Using technologies like Kafka, the project can store large amounts of data for extended periods, integrating with multiple technologies for comprehensive data management.

## Camel case integration layer.

This layer plays a pivotal role in integrating messaging protocols with the software, ensuring seamless communication and data flow.

The Bobby Car project, with its innovative use of technology and software, is a beacon in the automotive industry, demonstrating how connected and software-defined vehicles can significantly enhance operational efficiency and user experience in smart cities, smart factories, and beyond.

## Efficient real-time data processing and development framework

An essential aspect of Bobby Car's approach is the real-time management and processing of data. The system operates within the OpenShift platform, where events are stored in memory for quick access and faster processing. This setup allows for efficient data handling, essential for applications like connected cars, where GPS coordinates, metrics, and zone changes are continuously sent to the cloud. The cloud processes this data, and the results are displayed on a dashboard, providing real-time insights necessary for decision-making or AI integration.

Bobby Car has implemented a unique software development framework called 'Tablespaces.' This approach allows developers to work without installing dependencies on their devices. They use a browser-based integrated development environment (IDE), facilitating